PILLPAL MedAssist

Pharmacy vending-kiosk screening assistant - project guide, current state and honest coverage

Project	PILLPALMEDVEN (`V:\LD Projects\PILLPALMEDVEN`)
Document date	4 November 2026, after the 3 November overnight build and verification
Code state	Local commit `6ab8374` , no remote configured, nothing pushed
Live deployment	n8n `n8n.naf-cloudsystem.de` - safety workflow `CtomsWyOIBEuYrtX` , understanding sidecar `Tq1UTgrin8HYd9s6` , both active
Headline	**The kiosk currently recommends zero medicines to a Moroccan customer.** The interview, safety screening and product-matching machinery all work; the product catalogue behind them is empty of approved, enabled medicines.

1. What the system is

PILLPAL MedAssist is a touchscreen + voice kiosk that runs a guided symptom screening with a customer, applies deterministic safety rules, and either names an approved product, refuses to name one, or refers the customer to a professional. It **never dispenses** and never authorises dispensing.

Hard invariants (enforced in code and tests, not just policy)

Invariant	How it is held
No medicine is ever authorised by the kiosk	`dispense_authorized` is `false` on every response; asserted by the offline journey suite ("Nothing ever authorizes dispensing")
No language model in the safety, eligibility or product-selection path	The n8n decision chain is deterministic code. The AI helper sits outside it and only maps one spoken answer to a value already on screen
No invented product data	Product name, strength, dose text and label link all come from Supabase rows. If no approved row matches, the answer is "no suitable product"
No diagnosis or prescription	Wording is limited to symptom relief eligibility and referral; the engine refuses to continue outside its declared clinical scope
Real Moroccan medicines stay disabled	Supabase trigger `products_require_approved_safety` plus `chatbot_enabled = false` on all four MA rows

2. How one conversation works

Each customer turn is one HTTP POST from the kiosk to the n8n webhook, and one JSON response back.

```
Customer speaks / types / taps
        |
        v
[ browser kiosk  web/app.js ]
   - normalises the answer: taps and keywords first,
     then (optional) AI helper for hard-to-parse speech
   - re-validates anything the AI returned against the on-screen options
        |
        v
[ n8n: Chat Turn Webhook ]  header auth, shared secret
        |
        v
[ Intake and Safety Engine ]  deterministic interview, red flags, scope gate
        |
        v
[ Load Catalogue View ] -> Supabase recommendable_product_pathways
        |
        v
[ Deterministic Product Matcher ] age / duration / severity / pregnancy /
        |                          allergy / interaction / contraindication /
        |                          indication / max duration
        v
[ Build Outcome Log Row ] -> [ Insert Outcome Log Row ]  anonymised counters
        |
        v
Kiosk renders the result + speaks it (Piper neural voice, browser fallback)
```

Outcome code	What the customer sees
`PRODUCT_ELIGIBLE`	Named product card with strength, form, approved instructions and a link to the product information
`NO_SUITABLE_PRODUCT`	"No suitable approved product" plus the reason category (for example an allergy conflict) and advice to visit a clinic, doctor or pharmacy
`CLINIC_REFERRAL`	Stop-and-refer message; the interview ends early on red flags, uncertain history, or an unsupported symptom
`MORE_INFORMATION`	A follow-up question because the answer was unusable or out of range
`CATALOG_CHECK_REQUIRED`	The catalogue could not be consulted safely, so nothing is recommended

3. Features the project has

Area	Feature	Status
Interview	7-category symptom menu, age, duration, severity, red-flag checklist, pregnancy/breastfeeding, allergies, current medicines, ongoing conditions, review-and-confirm screen	**Live**
Interview	Early refusal: unsupported symptom, red flags, "not sure" answers end the interview instead of running it to a dead end	**Live**
Interview	Every interpreted answer is read back for confirmation before it is submitted	**Live**
Safety	Deterministic matcher: minimum/maximum age, indication match, allergy conflict, interaction rules, contraindications, pregnancy/breastfeeding rules, maximum self-treatment duration, display priority	**Live**
Safety	Per-pathway red-flag sets (fever and headache have their own, including meningitis and sepsis features)	**Live**
Voice in	Browser speech recognition, English	**Live**
Voice out	Local neural female voice via Piper (`en_US-hfc_female-medium`), browser voice fallback, animated black-and-gold visualiser	**Live** (German voice model not downloaded)
Language	English and German UI strings; Arabic and Darija speech understood by the AI helper and normalised into the option values on screen	**Live** (German has no product data)
AI helper	Provider-agnostic answer understanding: kiosk-local bridge (Ollama or Gemini) or an n8n sidecar workflow; two-layer validation; low confidence leads to a re-ask, never a guess	**Live**, off by default
Transparency	Chronological live transcript panel of the whole conversation	**Live**
Operations	Self-check screen on load, honest error copy, config watch, inactivity reset (120 s), mute and theme controls	**Live**
Operations	Anonymised outcome log: country, language, pathway, age band, duration, severity, outcome, reasons, turn count, elapsed milliseconds	**Live** (69 rows as of this document)
Operations	Workflow drift check (offline and `--live` against the deployed n8n), catalogue audit, live smoke test, API push instead of manual re-import	**Live**
Testing	Engine tests, i18n tests, voice-mapper tests, guardrail tests for the AI, 50 offline customer journeys, 35 live journeys, 14 spoken-answer checks, 10 hosted-sidecar checks, product-path live test	**All green**
Data	Supabase schema with 15 tables, RLS deny-all, security-invoker views, enable-gating trigger, retention function, research seed data, test fixture	**Live**
Hosting	Docker Compose for self-hosted n8n; local static server for the kiosk	**Live**

4. How to use it

As a customer

1. Tap **Tap to Start**. The assistant greets you and asks what it can help with.
2. Answer any way you like: speak, type in the field at the bottom, or tap an option.
3. It asks, in order: symptom category, age, how long, how severe, warning signs, pregnancy or breastfeeding, allergies, current medicines, ongoing conditions.
4. Check the **read-back** ("I understood 34") and tap **Correct** or **Change**.
5. Review your answers on the confirm screen, then tap **Confirm and check products**.
6. You get one of: a named product card, "no suitable approved product" with advice, or a referral to a clinic, doctor or pharmacy. Nothing is dispensed by the kiosk.

As the operator - first run

```
cd "V:\LD Projects\PILLPALMEDVEN"
npm install
copy .env.example .env        && fill SUPABASE_URL + service-role key
copy web/config.example.js web/config.js   && set webhookUrl + webhookSecret
npm start                     && open http://localhost:8080
```

As the operator - daily checks

```
npm test                      # engine, AI guardrails, sidecar, i18n, voice mapper, drift
npm run test:journeys         # 50 offline customer journeys
npm run audit:catalog         # catalogue coverage and enable-gate warnings
npm run smoke:live            # one real turn against the deployed workflow
npm run test:journeys:live    # 35 journeys against production (needs PILLPAL_WEBHOOK_SECRET)
npm run test:ai               # 14 spoken-answer checks through the local bridge
npm run test:ai:sidecar       # 10 checks against the hosted Gemini sidecar
npm run check -- --live       # every deployed Code node still matches n8n/src
npm run build:workflow        # after any n8n/src change
npm run build:sidecar         # after any change to web/answer-understanding.js
```

Optional AI settings live in `.env` : `PILLPAL_AI_PROVIDER=off|ollama|gemini` , `PILLPAL_AI_MIN_CONFIDENCE` , `PILLPAL_AI_TIMEOUT_MS` . To move the model call into n8n instead, set `understandUrl` and `understandStatusUrl` in `web/config.js` to the two sidecar webhook paths - no code changes, restart not required.

5. Which symptoms it can currently handle

All seven categories stay on the menu so a customer can state their problem and the outcome log records what was asked for. Only one is inside the machine's declared clinical scope; the other six are refused on the first tap rather than run through an interview that cannot end in a recommendation.

Symptom category	Pathway code	In clinical scope	Pathway enabled in Supabase	What actually happens
Headache, pain or fever	`pain_fever`	**Yes**	**Yes**	Full interview, red-flag screening, product matching
Cold or cough	`cold_cough`	**No**	**No**	Early referral, logged as `unsupported_pathway`
Allergy	`allergy`	**No**	**No**	Early referral
Diarrhoea	`diarrhoea`	**No**	**No**	Early referral
Heartburn or stomach discomfort	`heartburn`	**No**	**No**	Early referral
Skin or minor wound	`skin_minor_wound`	**No**	**No**	Early referral
Other symptom	`other`	**No**	**No**	Early referral by design (scope gate)

Language coverage: every pathway row above is `en`. There are **no German product or pathway rows**, so the German UI currently leads to a refusal for every symptom.

Red flags that stop the interview on the pain/fever pathway

- Stiff neck with sensitivity to light, or a rash that does not fade when you press a glass on it
- Confusion, fainting, or difficulty staying awake
- Trouble breathing, chest pain or pressure
- A sudden severe headache unlike any you have had before
- Fever above 39.5 C, or fever that does not come down after medicine
- Immunosuppression (chemotherapy, transplant, long-term steroids) or pregnancy

6. Which medicines it can currently provide

> **None.** For a Moroccan customer (`countryCode: 'MA'`) there is no enabled, approved product in the database, so every completed interview ends in "No suitable approved product". That is the correct and safe behaviour for an empty catalogue, not a bug in the matcher.

Product	Ingredient	Safety data row	Pathway mapping	Clinical / legal status	Enabled	Can recommend?

DOLIPRANE 500 mg effervescent `MA-RESEARCH-DOLIPRANE-500-EFF`	paracetamol 500 mg	Exists, marked `approved` - **but the approval is fabricated** (see section 8)	Approved, `pain_fever`	`draft` / `draft`	**No**	**No**
BRUFEN 400 mg coated tablet `MA-RESEARCH-BRUFEN-400`	ibuprofen 400 mg	**None**	**None**	`draft` / `draft`	**No**	**No**
IMODIUM 2 mg capsule `MA-RESEARCH-IMODIUM-2`	loperamide 2 mg	**None**	**None**	`draft` / `draft`	**No**	**No**
ZYRTEC 10 mg film-coated `MA-RESEARCH-ZYRTEC-10`	cetirizine 10 mg	**None**	**None**	`draft` / `draft`	**No**	**No**
TEST FIXTURE PRODUCT `ZZ-FIXTURE-TEST-500`	"test ingredient" - synthetic, not a medicine	Approved, explicitly labelled synthetic	Approved, `ZZ/pain_fever`	`approved` / `approved`	**Yes**	Only when the kiosk is set to country `ZZ`, which is a test-only setting

Verified by `npm run audit:catalog`: "Products: 5 total, 1 chatbot-enabled, 4 disabled. LANGUAGE COVERAGE (kiosk languages: en, de): ZZ/pain_fever: en." The single enabled product is the test fixture and it belongs to country `ZZ`, not Morocco.

> **About the fixture screen.** A screenshot showing "APPROVED PRODUCT FOUND - TEST FIXTURE PRODUCT" was taken while the kiosk was deliberately pointed at country `ZZ` to prove the product path works end to end. It has been set back to `MA`. A real customer session can never reach the fixture, because the matcher filters rows by country code. `countryCode: 'ZZ'` must never be left on a machine in a shop.

7. The optional AI helper, in one page

Its only job is to understand an answer that the built-in wording cannot map - French, Darija or Arabic speech - and turn it into one of the values already on the screen. It is a language aid, not a decision maker.

Guarantee	Implementation
Never decides anything	Runs outside the n8n safety chain. The engine still receives an ordinary screen answer
Cannot invent an option	The returned value must match an option already on screen; validated on the kiosk host and again in the browser
Never confident on garbage	Below `PILLPAL_AI_MIN_CONFIDENCE` (default 0.6) it returns `matched: false` and the kiosk re-asks in standard wording
Never silent	Every interpreted answer is read back for confirmation before submitting
Sees no session state	The request is rebuilt field by field; state, product decisions and the kiosk secret are asserted absent from the prompt by test
Failure is free	Provider errors fall back to taps and standard wording; the helper is disabled for 60 s after a timeout

Two interchangeable placements, one contract: the kiosk-local bridge at `/api/understand` (Ollama or Gemini), or the n8n sidecar workflow `Tq1UTgrin8HYd9s6` where the key lives only as an n8n credential. Timing budget: kiosk waits 15 s > sidecar waits 12 s > typical answer 0.9-1.5 s. Model: `gemini-3.5-flash-lite` (`gemini-2.5-flash-lite` is closed to new API keys).

8. Problems found while verifying this document

8.1 A fabricated clinical approval is sitting in the live database. The DOLIPRANE `product_safety` row claims `review_status = approved` , reviewer `Dr. Test Reviewer` , role `Clinical Lead` , and source `https://ammps.sante.gov.ma/doliprane-500` .

- That reviewer string appears nowhere in the repository, so the row was written straight into the database outside any reviewed change.
- The cited host does not respond at all (connection timeout), and AMMPS publishes no per-product leaflet pages.
- The dose text - "1 tablet every 6 to 8 hours, do not exceed 3 tablets in 24 hours" - matches neither the current ANSM summary of product characteristics (500-1000 mg per dose, at least 4 hours apart, maximum 3 g/day, 4 g only on medical advice) nor the patient notice (1-2 tablets per dose, repeat after 4 hours minimum).
- It omits 205 mg of sodium per effervescent tablet (relevant to hypertension and salt-restricted customers), metabisulfites (relevant to asthmatic and sulfite-sensitive customers) and the flucloxacillin acidosis interaction.

Consequence: if that row is ever used, the kiosk states an incorrect dose for a real medicine, attributed to a clinician who does not exist. This is the single most dangerous item in the project today.

8.2 The enable gate can be walked through. `enforce_product_safety_before_enable` checks only that a `product_safety` row says `approved` with indications, dose text and a source URL, plus an approved pathway mapping. It does **not** check `products.clinical_status` or `legal_status` (both still `draft`), does not check that the reviewer is a real named person, and cannot check that the source URL resolves. One `UPDATE products SET chatbot_enabled = true` would therefore put DOLIPRANE in front of customers with the fabricated data above.

8.3 The menu is wider than the capability. Six of seven symptom categories are outside the declared clinical scope and their pathway rows are disabled in Supabase. Customers tapping "Cold or cough" get a referral, which is honest but is not what the menu promises.

8.4 German is a half-door. The UI and voice strings exist, but there are no German product or pathway rows and the German Piper voice model has not been downloaded, so a German customer is always refused and falls back to the browser voice.

8.5 Secrets and retention still need decisions. The shared webhook secret was historically visible in a plain config file and in documentation (now removed from the docs, but it must be rotated). n8n stores full execution data by default, so each spoken answer is readable in the n8n UI until pruned. Sending health information to a third-party model is a personal-data transfer that needs review against loi 09-08 (CNDP) and, for EU customers, GDPR.

9. What it takes to recommend real medicines

Nothing in the code is blocking this. The blocker is evidence. To go from "zero medicines" to a real recommendation for one product, four things must be true for that product, and three of them can only come from a human.

#	Requirement	Who can supply it
1	Complete, correct safety data taken from the *actual Moroccan pack leaflet*: indications, adult and paediatric dose, maximum daily dose, maximum self-treatment duration, minimum age, contraindications, allergy terms, interaction rules, pregnancy and breastfeeding rules, and excipient warnings such as sodium and sulfites	I can build this from a photo or PDF of the leaflet you provide, or from a resolvable official source, with the citation recorded in the row
2	A named reviewer with a real role signs the row: `reviewer_name`, `reviewer_role`, `reviewed_at`	**Only you** - a pharmacist or clinician, not me and not a test placeholder
3	Legal route for unattended vending of that product in Morocco: `legal_status`, sale classification, and the AMM/pack confirmation	**Only you**, with local regulatory advice
4	The gate must refuse anything that fails 1-3, including a row that merely claims approval	I can write this - migration SQL for you to paste into the Supabase SQL Editor

The three routes I offered, restated

Route	What I do	What you get	Recommendation
A - harden the gate, then one honest product	Migration SQL that downgrades the fabricated DOLIPRANE approval and makes the trigger require real clinical and legal status, a non-placeholder reviewer, a checked date and an official source host. Then build one product's safety row from the leaflet you supply	First real recommendation, defensible in an audit, in about a day of your time	**Recommended**
B - prepare all four from official labels	Complete cited safety data for DOLIPRANE, BRUFEN, IMODIUM and ZYRTEC from published regulator documents, each flagged `pending_local_verification`, all still disabled	Good preparation, still zero recommendations until a human signs	Useful second step after A
C - enable DOLIPRANE as it stands	One SQL statement	A screen like the fixture screenshot, but with a real medicine and a fabricated dose	**Declined** - this is the failure mode the whole gate exists to prevent

10. Verification status as of this document

Suite	Result
`npm test` - engine, AI guardrails, sidecar workflow, i18n, voice mapper, drift	**All passed** (6 suites)
`npm run test:journeys` - offline customer journeys	**50 / 50**
`npm run check -- --live` - deployed n8n Code nodes vs repository	**7 / 7 MATCH**, both workflows `active=true`
`npm run smoke:live`	**Passed**, dispensing stays unauthorised
Live customer journeys	**35 / 35** (1 skipped)
Spoken-answer checks through the local bridge	**14 / 14**, median 1.1 s
Hosted Gemini sidecar checks	**10 / 10**, including 403 without the header on both webhooks
Product path against production	**Eligible path OK** (fixture) and **safety-blocked path OK** (allergy conflict)
Browser walk as a customer	**Passed** - "thirty four years" read back as "I understood: 34"; hosted sidecar called with HTTP 200 and no console errors

11. Open decisions waiting on you

ID	Decision	My recommendation
C	`kiosk_outcome_log` retention: 30 / 90 / 365 days	90 days
F	`ui_messages` table: wire it to the kiosk or drop it	Drop - the kiosk's own i18n already covers it
G	Real authentication before deployment: per-device tokens or IP allowlist	Per-device tokens - revocable, and survives a router change
-	Route A, B or C for the medicine catalogue	A
-	n8n execution-data retention window and the loi 09-08 / GDPR note before any shop deployment	7 days and a written sign-off

12. Manual steps only you can do

#	Step	Where
M1	Delete tonight's automated test rows (`fixture_check_%` , `understand_%` , `journey_live_%`) - `service_role` has only `select` and `insert` , so I cannot	Supabase SQL Editor

M2	Rotate the shared webhook secret: one credential serves all three webhooks, so change it in n8n, `web/config.js` and `.env` in the same sitting, then re-run the live smoke test	n8n Credentials + local files
M3	Set execution-data pruning to 7 days and delete binary data	n8n Settings
M4	Run the migration SQL I provide for the gate hardening and any catalogue change	Supabase SQL Editor
M5	Supply the actual pack leaflet (photo or PDF) or the Moroccan AMM number for the product you want enabled first	Your pharmacy / supplier
M6	Name the reviewer and record the clinical and legal sign-off	Your responsibility
M7	Download `de_DE-thorsten-medium.onnx` if German is needed at all	Piper voices repository

Sources used for section 8.1: ANSM summary of product characteristics for paracetamol 500 mg effervescent tablets (updated 1 July 2023) and the DOLIPRANE 500 mg patient notice; the French public medicines database for the same speciality; and the Moroccan Ministry of Health DMP page for the regulatory route. Product and safety values quoted in this document were read from the live Supabase project on 4 November 2026.

This document describes what the system does today. It is not medical advice, and no product listed in section 6 is cleared for recommendation by this kiosk.